

LANDING1.ANL

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*
*          STAAD.Pro
*          Version 2007    Build 02
*          Proprietary Program of
*          Research Engineers, Intl.
*          Date=    NOV 24, 2009
*          Time=    6:29:14
*
*          USER ID: WS Atkins and Partners Overseas
*****
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INPUT FILE: LANDING1.STD

```
1. STAAD SPACE BLDG 011
2. START JOB INFORMATION
3. ENGINEER DATE 02-FEB-09
4. JOB NAME D.C.C. (BUILDING NO.11)-DCP4
5. *--LANDING SLAB IN PLANE FORCES
6. END JOB INFORMATION
7. INPUT WIDTH 79
8. UNIT METER KN
9. JOINT COORDINATES
10. 3655 9.14275 16 -1.15147; 3827 8.66485 16 -3.50207; 3901 7.4576 16 -3.25317
11. 3902 9.8721 16 -3.75097; 3916 7.9355 16 -0.90257; 3917 10.35 16 -1.40037
12. 4813 10.3102 16 -1.59625; 4814 10.109 16 -1.55477; 4815 10.1488 16 -1.35889
13. 4816 10.2704 16 -1.79214; 4817 10.0691 16 -1.75065; 4818 10.2305 16 -1.98802
14. 4819 10.0293 16 -1.94654; 4820 10.1907 16 -2.1839; 4821 9.98949 16 -2.14242
15. 4822 10.1509 16 -2.37979; 4823 9.94967 16 -2.3383; 4824 10.111 16 -2.57567
16. 4825 9.90984 16 -2.53419; 4826 10.0712 16 -2.77155; 4827 9.87002 16 -2.73007
17. 4828 10.0314 16 -2.96744; 4829 9.83019 16 -2.92595; 4830 9.99158 16 -3.16332
18. 4831 9.79037 16 -3.12184; 4832 9.95175 16 -3.3592; 4833 9.75054 16 -3.31772
19. 4834 9.91193 16 -3.55509; 4835 9.71072 16 -3.5136; 4836 9.67089 16 -3.70949
20. 4837 9.90776 16 -1.51329; 4838 9.94758 16 -1.3174; 4839 9.86793 16 -1.70917
21. 4840 9.82811 16 -1.90505; 4841 9.78828 16 -2.10094; 4842 9.74846 16 -2.29682
22. 4843 9.70863 16 -2.4927; 4844 9.66881 16 -2.68859; 4845 9.62898 16 -2.88447
23. 4846 9.58916 16 -3.08035; 4847 9.54933 16 -3.27624; 4848 9.50951 16 -3.47212
24. 4849 9.46968 16 -3.668; 4850 9.70655 16 -1.4718; 4851 9.74638 16 -1.27592
25. 4852 9.66673 16 -1.66769; 4853 9.6269 16 -1.86357; 4854 9.58708 16 -2.05945
26. 4855 9.54725 16 -2.25534; 4856 9.50742 16 -2.45122; 4857 9.4676 16 -2.6471
27. 4858 9.42777 16 -2.84299; 4859 9.38795 16 -3.03887; 4860 9.34812 16 -3.23475
28. 4861 9.3083 16 -3.43064; 4862 9.26847 16 -3.62652; 4863 9.50534 16 -1.43032
29. 4864 9.54517 16 -1.23444; 4865 9.46552 16 -1.6262; 4866 9.42569 16 -1.82209
30. 4867 9.38587 16 -2.01797; 4868 9.34604 16 -2.21385; 4869 9.30622 16 -2.40974
31. 4870 9.26639 16 -2.60562; 4871 9.22657 16 -2.8015; 4872 9.18674 16 -2.99739
32. 4873 9.14692 16 -3.19327; 4874 9.10709 16 -3.38915; 4875 9.06727 16 -3.58504
33. 4876 9.30413 16 -1.38884; 4877 9.34396 16 -1.19295; 4878 9.26431 16 -1.58472
34. 4879 9.22448 16 -1.7806; 4880 9.18466 16 -1.97649; 4881 9.14483 16 -2.17237
35. 4882 9.10501 16 -2.36825; 4883 9.06518 16 -2.56414; 4884 9.02536 16 -2.76002
36. 4885 8.98553 16 -2.9559; 4886 8.94571 16 -3.15179; 4887 8.90588 16 -3.34767
37. 4888 8.86606 16 -3.54355; 4889 9.10293 16 -1.34735; 4890 9.0631 16 -1.54324
38. 4891 9.02328 16 -1.73912; 4892 8.98345 16 -1.935; 4893 8.94362 16 -2.13089
39. 4894 8.9038 16 -2.32677; 4895 8.86397 16 -2.52265; 4896 8.82415 16 -2.71854
40. 4897 8.78432 16 -2.91442; 4898 8.7445 16 -3.1103; 4899 8.70467 16 -3.30619
    BLDG 011
41. 4900 8.90172 16 -1.30587; 4901 8.94154 16 -1.10999; 4902 8.86189 16 -1.50175
42. 4903 8.82207 16 -1.69764; 4904 8.78224 16 -1.89352; 4905 8.74242 16 -2.0894
43. 4906 8.70259 16 -2.28529; 4907 8.66277 16 -2.48117; 4908 8.62294 16 -2.67705
44. 4909 8.58312 16 -2.87294; 4910 8.54329 16 -3.06882; 4911 8.50347 16 -3.2647
45. 4912 8.46364 16 -3.46059; 4913 8.70051 16 -1.26439; 4914 8.74033 16 -1.0685
46. 4915 8.66068 16 -1.46027; 4916 8.62086 16 -1.65615; 4917 8.58103 16 -1.85204
47. 4918 8.54121 16 -2.04792; 4919 8.50138 16 -2.2438; 4920 8.46156 16 -2.43969
48. 4921 8.42173 16 -2.63557; 4922 8.38191 16 -2.83145; 4923 8.34208 16 -3.02734
49. 4924 8.30226 16 -3.22322; 4925 8.26243 16 -3.4191; 4926 8.4993 16 -1.2229
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51. 4930 8.37983 16 -1.81055; 4931 8.34 16 -2.00644; 4932 8.30018 16 -2.20232
52. 4933 8.26035 16 -2.3982; 4934 8.22052 16 -2.59409; 4935 8.1807 16 -2.78997
53. 4936 8.14087 16 -2.98585; 4937 8.10105 16 -3.18174; 4938 8.06122 16 -3.37762
54. 4939 8.29809 16 -1.18142; 4940 8.33792 16 -0.985537; 4941 8.25827 16 -1.3773
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*LANDING SLAB
IN-PLANE STRESS
ANALYSIS*

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55. 4942 8.21844 16 -1.57319; 4943 8.17862 16 -1.76907; 4944 8.13879 16 -1.96495
 56. 4945 8.09897 16 -2.16084; 4946 8.05914 16 -2.35672; 4947 8.01932 16 -2.5526
 57. 4948 7.97949 16 -2.74849; 4949 7.93967 16 -2.94437; 4950 7.89984 16 -3.14025
 58. 4951 7.86002 16 -3.33614; 4952 8.09688 16 -1.13994; 4953 8.13671 16 -0.944053
 59. 4954 8.05706 16 -1.33582; 4955 8.01723 16 -1.5317; 4956 7.97741 16 -1.72759
 60. 4957 7.93758 16 -1.92347; 4958 7.89776 16 -2.11935; 4959 7.85793 16 -2.31524
 61. 4960 7.81811 16 -2.51112; 4961 7.77828 16 -2.707; 4962 7.73846 16 -2.90289
 62. 4963 7.69863 16 -3.09877; 4964 7.65881 16 -3.29465; 4965 7.89568 16 -1.09845
 63. 4966 7.85585 16 -1.29434; 4967 7.81603 16 -1.49022; 4968 7.7762 16 -1.6861
 64. 4969 7.73638 16 -1.88199; 4970 7.69655 16 -2.07787; 4971 7.65673 16 -2.27375
 65. 4972 7.6169 16 -2.46964; 4973 7.57708 16 -2.66552; 4974 7.53725 16 -2.8614
 66. 4975 7.49743 16 -3.05729; 4977 9.39551 16 -3.81336

67. MEMBER INCIDENCES

68. 7941 4886 4897; 7943 4886 4977; 7944 3655 4889; 7945 4889 4890; 7946 4890 4891
 69. 7947 4891 4892; 7948 4892 4893; 7949 4893 4894; 7950 4894 4895; 7951 4895 4896
 70. 7952 4896 4897; 7953 4897 4898; 7954 4898 4899; 7955 4899 3827

71. *****

72. ELEMENT INCIDENCES SHELL

73. 7782 3917 4813 4814 4815; 7783 4813 4816 4817 4814; 7784 4816 4818 4819 4817
 74. 7785 4818 4820 4821 4819; 7786 4820 4822 4823 4821; 7787 4822 4824 4825 4823
 75. 7788 4824 4826 4827 4825; 7789 4826 4828 4829 4827; 7790 4828 4830 4831 4829
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 83. 7812 4843 4844 4857 4856; 7813 4844 4845 4858 4857; 7814 4845 4846 4859 4858
 84. 7815 4846 4847 4860 4859; 7816 4847 4848 4861 4860; 7817 4848 4849 4862 4861
 85. 7818 4851 4850 4863 4864; 7819 4850 4852 4865 4863; 7820 4852 4853 4866 4865
 86. 7821 4853 4854 4867 4866; 7822 4854 4855 4868 4867; 7823 4855 4856 4869 4868
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97. 7856 3655 4889 4900 4901; 7857 4889 4890 4902 4900; 7858 4890 4891 4903 4902
 98. 7859 4891 4892 4904 4903; 7860 4892 4893 4905 4904; 7861 4893 4894 4906 4905
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 120. 7935 4961 4962 4974 4973; 7937 4962 4963 4975 4974; 7938 4963 4964 3901 4975
 121. DEFINE MATERIAL START
 122. ISOTROPIC CONCRETE
 123. E 3.16E+007
 124. POISSON 0.17
 125. DENSITY 24
 126. ALPHA 1E-005
 127. DAMP 0.05

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128. ISOTROPIC MATERIAL
 129. E 3.16E+007
 130. POISSON 0.17
 131. DENSITY 0
 132. ALPHA 1E-005
 133. DAMP 0.05
 134. ISOTROPIC STEEL
 135. E 2.05E+008
 136. POISSON 0.3
 137. DENSITY 76.8195
 138. ALPHA 1.2E-005
 139. DAMP 0.03
 140. END DEFINE MATERIAL
 141. ELEMENT PROPERTY
 142. 7782 TO 7853 7856 TO 7915 7917 7919 7921 7923 7925 7927 7929 7931 7933 7935 -
 143. 7937 7938 THICKNESS 0.2
 144. MEMBER PROPERTY AMERICAN
 145. 7941 7943 PRIS YD 0.032
 146. 7944 TO 7955 PRIS YD 0.8 ZD 0.3
 147. CONSTANTS
 148. MATERIAL CONCRETE MEMB 7782 TO 7853 7856 TO 7915 7917 7919 7921 7923 7925 -
 149. 7927 7929 7931 7933 7935 7937 7938 7944 TO 7955
 150. MATERIAL STEEL MEMB 7941 7943
 151. SUPPORTS
 152. 3655 4945 4958 PINNED
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153. LOAD 11 WIND LOAD 1 (0 DEG ,INTERNAL SUCTION ,CPI=-0.3)
 154. ** SPIDER FRAME LOAD (RIGHT WING - STAIR LANDING)
 155. MEMBER LOAD
 156. 7943 UNI X 275
 157. JOINT LOAD
 158. 4832 FZ 73
 159. LOAD 15 WIND LOAD 5 (0 DEG ,INTERNAL SUCTION ,CPI=-0.3)
 160. ** SPIDER FRAME LOAD (RIGHT WING - STAIR LANDING)
 161. JOINT LOAD
 162. 4832 FZ -96
 163. 4832 FX 165
 164. LOAD COMB 101
 165. 11 1.4
 166. LOAD COMB 102
 167. 15 1.4
 168. PERFORM ANALYSIS

PROBLEM STATISTICS

NUMBER OF JOINTS/MEMBER+ELEMENTS/SUPPORTS = 170/ 158/ 3
 ORIGINAL/FINAL BAND-WIDTH= 166/ 22/ 138 DOF
 TOTAL PRIMARY LOAD CASES = 2, TOTAL DEGREES OF FREEDOM = 1020
 SIZE OF STIFFNESS MATRIX = 141 DOUBLE KILO-WORDS
 REQD/AVAIL. DISK SPACE = 14.3/ >2000 MB, EXMEM = 1168.5 MB

169. LOAD LIST 101 102
 170. PRINT REACTIONS
 REACTION
 BLDG 011

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SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
3655	101	32.33	0.00	-84.76	0.00	0.00	0.00
	102	-25.68	0.00	-35.78	0.00	0.00	0.00
4945	101	-136.47	0.00	102.80	0.00	0.00	0.00

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